Integrated Autonomous System of Vehicle and Drone for Disaster Prevention
Drone Pose Optimization with Multi-Camera & UWB

Alpha project / Vertically Integrated Project 2023-2

APD

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Overview

What we did

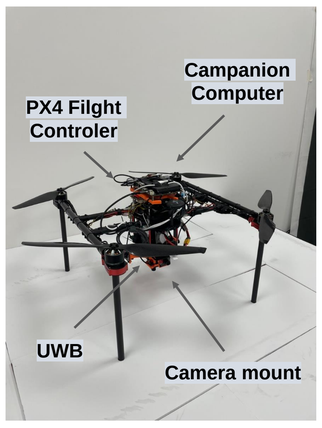
- Design multi-camera & UWB drone system
- Implement drone pose optimization algorithm
- Solve scale ambiguity & drift problem

Approach

- Stabilize drone from last semester and setting up our drone system
- SVO(Semi–Direct Monocular Visual Odometry) to get the drone pose
- Design Multi-camera & UWB system to optimize drone pose

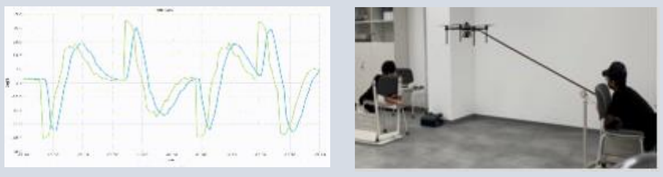
Design Drone

Design

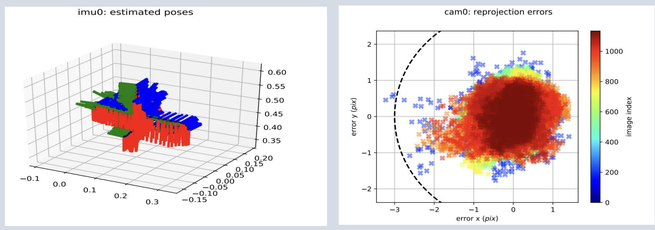


F660

PID tuning



Multi-camera & IMU calibration

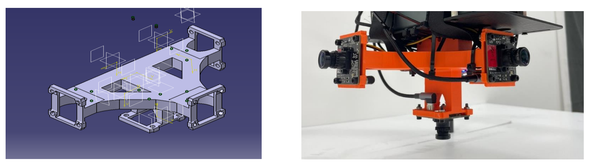


IMU position estimation | Camera-IMU reprojection error

Modeling & Drone

CATIA drone modeling | Our Drone

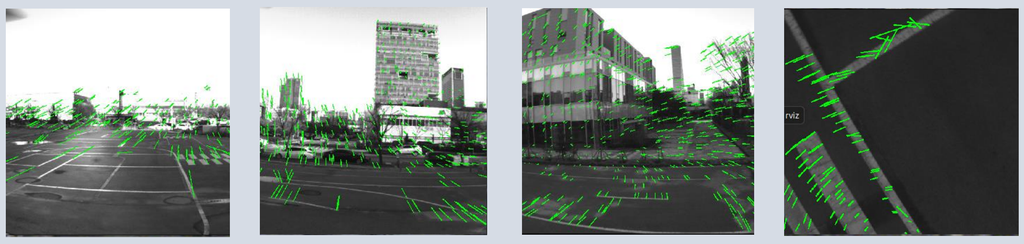
Multi-Camera



CATIA camera mount modeling | Camera mount

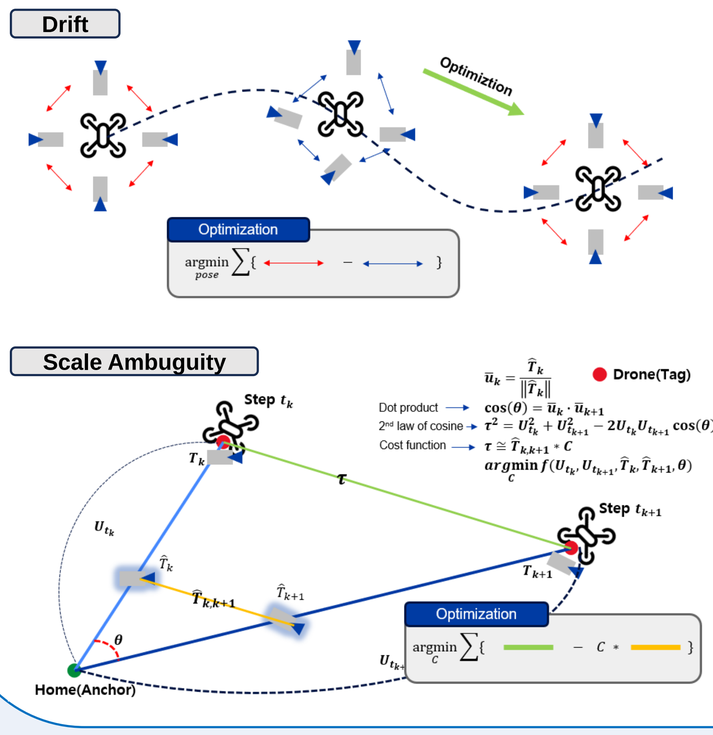
Using **multi-camera** we can optimize drone pose and can cover extensive area.

View of each Cameara



CAM1 view | CAM2 view | CAM3 view | CAM4 view

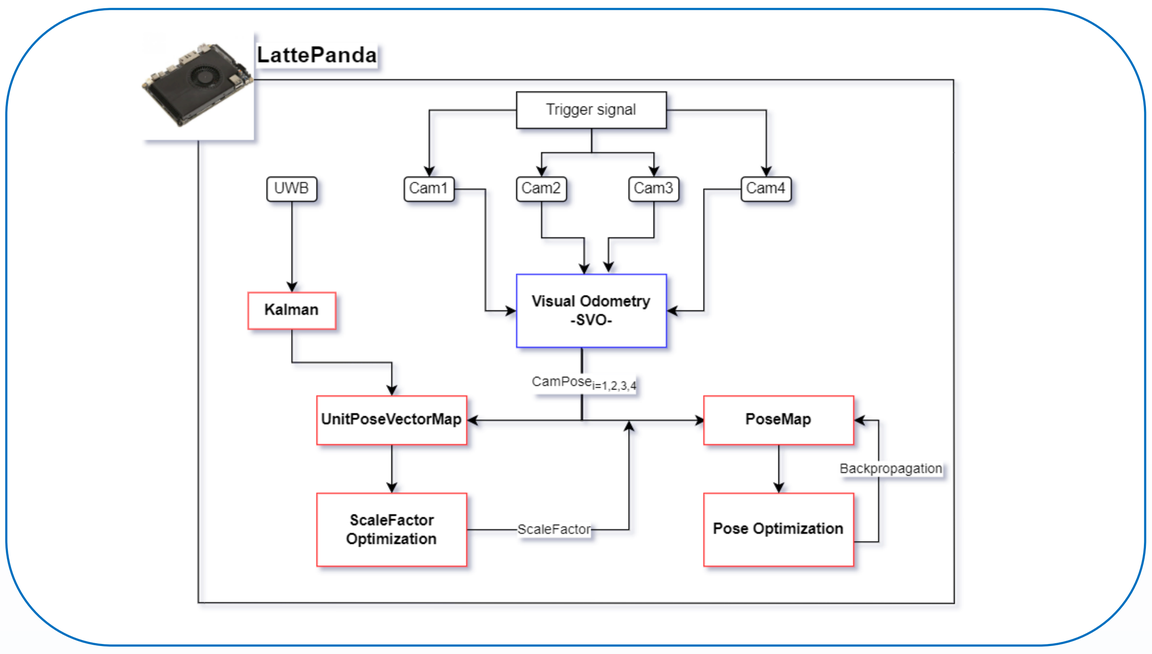
Pose Optimazation Using Multi-Camera & UWB



Each of the four cameras **optimizes each other's poses.**

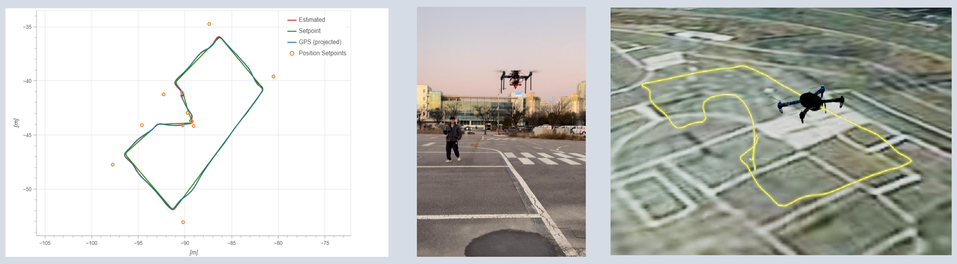
Scale problems that occur using monocular cameras are solved using **UWB**.

System Pipeline



Experiment

Outdoor Flight Test

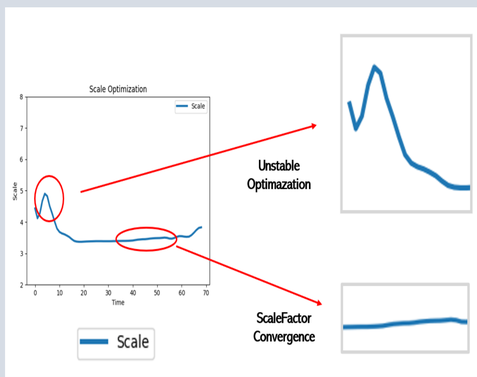


2D Estimated, GPS position | Outdoor filght test | 3D Estimated position

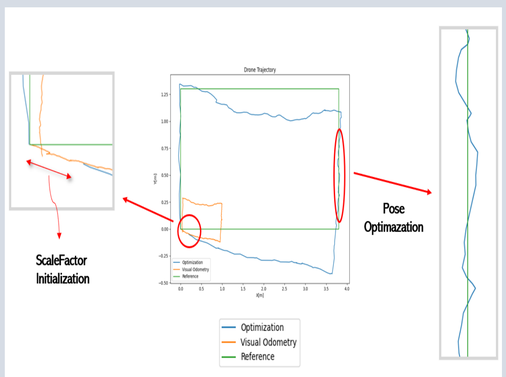
Indoor Algorithm Test

Experiment Enviroment | SVO to get drone pose

Algorithm Test Result

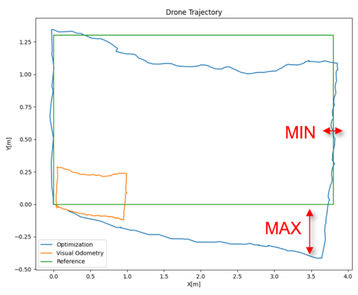


Algorithm Test Result



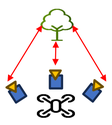
Conclusions

- **Design our drone system**
  Design multi-camera & UWB drone system that fly stably.
- **Implement drone pose optimization algorithm**



	Trajectory Error
MAX	0.415(m)
MIN	0.027(m)
RMSE	0.186(m)

Future Work



- **More Constraints for Pose Optimization**
  Additional constraints from each camera measurement, Better performance of Pose optimization

- **Wide Implementation UWB**
  When lots of smartphone have UWB, We can optimize each smartphone position from reference position

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